PowerScan PM9500 Two-Way Communications Tutorial

Introduction: This tutorial describes step by step the setup of the PowerScan PM9500 series scanners to use with the 2 way communication protocol. Per default, the scanner is working in 1 way. This means that you scan a barcode, the barcode is transmitted to the cradle, and the cradle transmits this data to the host.
Two Way protocol means that the host is sending a message back to the scanner after reception of the scanned data. Sending back messages to the Scanner can only be done when the cradle is connected to the host in USB-COM mode, RS232 mode or Ethernet.

The messages send back to the Scanner can be text messages to show on the display, pilot the LED's (Red & Green) on the scanner and pilot the beeper. It is not mandatory to have a display.

This tutorial does not describe programming of the two way protocol, but it gives an overview and tools to test and try the 2 way protocol.

Hardware & Software Setup : To test the articles and suggestions out of this tutorial require a minimum of next hardware.

a) Test via RS232
- PowerScan PM9500 (With or without display). Only the Mobile 433 Mhz / 910 Mhz versions allow 2 way. The Bluetooth variant doesn't support 2 way communications.
- Base Station (also called cradle). Model BC9030 or BC9160.
- RS232 cable.
- Power supply for the cradle.

b) Test via USB-COM
- PowerScan PM9500 (With or without display). Only the Mobile 433 Mhz / 910 Mhz versions allow 2 way. The Bluetooth variant doesn't support 2 way communications.
- Base Station (also called cradle). Model BC9030 or BC9160.
- USB cable.
- Power supply for the cradle (This is not mandatory, the cradle can be powered via the USB cable).
- USB-COM driver. This driver can be found on the Datalogic Website : http://www.datalogic.com/eng/support-services/automatic-data-capture/downloads/software-utilities-sw-2.html

c) Test via Ethernet
- PowerScan PM9500 (With or without display). Only the Mobile 433 Mhz / 910 Mhz versions allow 2 way. The Bluetooth variant doesn't support 2 way communications.
- Base Station (also called cradle). Model BC9180.
- Ethernet network cable.
- Power supply for the cradle.

The hardware setup is described in the reference manual of the PowerScan PM9500:
http://www.datalogic.com/tools/download.aspx?iddwnfile=10678&path=%2fupload%2fmarketlit%2fmanuals%2fpowerscan%2f820053614.pdf&name=820053614

First Simple 1- way test:

a) Setup Scanner: The below barcodes & procedures is to setup the scanner from scratch and do a basic 1 way test. Scan the appropriate barcodes and steps described below. If for one or another reason the scanner does not scans and the LED is blinking, please hold the trigger for 10 seconds to reset the scanner.

Step 1: **Unlink**: this barcode is used to remove the link between the scanner and previous connected cradle

Step 2: **Link**: Put the scanner in the cradle to link cradle and scanner.

Step 3: **Restore Factory Default**: restore all settings to factory default.

Step 4: **Interface Selection**: depending on the cable and interface you want to use, scan 1 of the 3 below barcodes.

RS232 Interface	USB-COM Interface	Ethernet Interface

Step 5: Test the current setting.

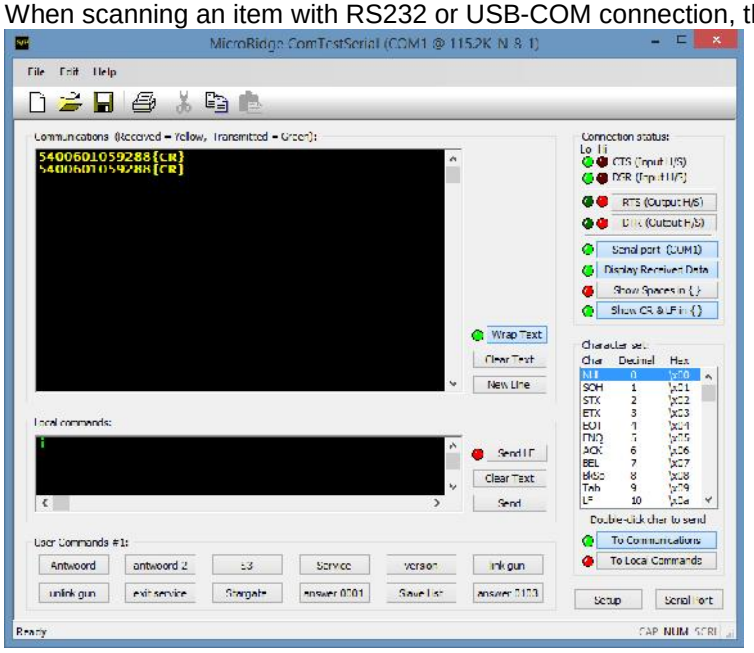
For RS232 and USB-COM interface, you need to test with a serial port terminal tool. ComTestSerial is a freeware terminal tool that can be downloaded at the next link:
http://www.microridge.com/ftp/comtestserial/Install_ComTestSerial_3_0_0_113.exe

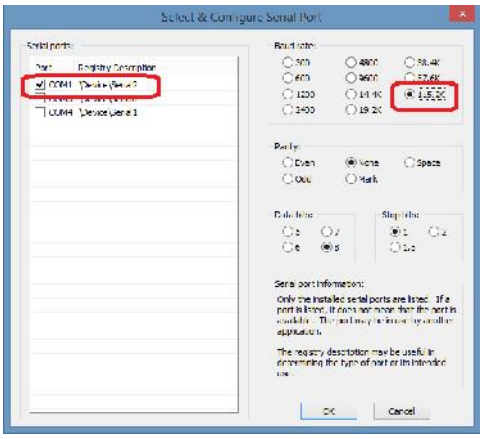
Download and install this tool. Select the correct serial port and set the baud rate to 115200. Per default, RS232 and USB-COM speed of the cradle is set to 115200

For Ethernet, you need Aladdin (minimal version 1.7) to detect the IP address of the cradle; and you can download the freeware Hercules to use as an terminal tool :
http://www.hw-group.com/products/hercules/index_en.html
http://new.hwg.cz/files/download/sw/version/hercules_3-2-8.exe

Some print screens of the tools:

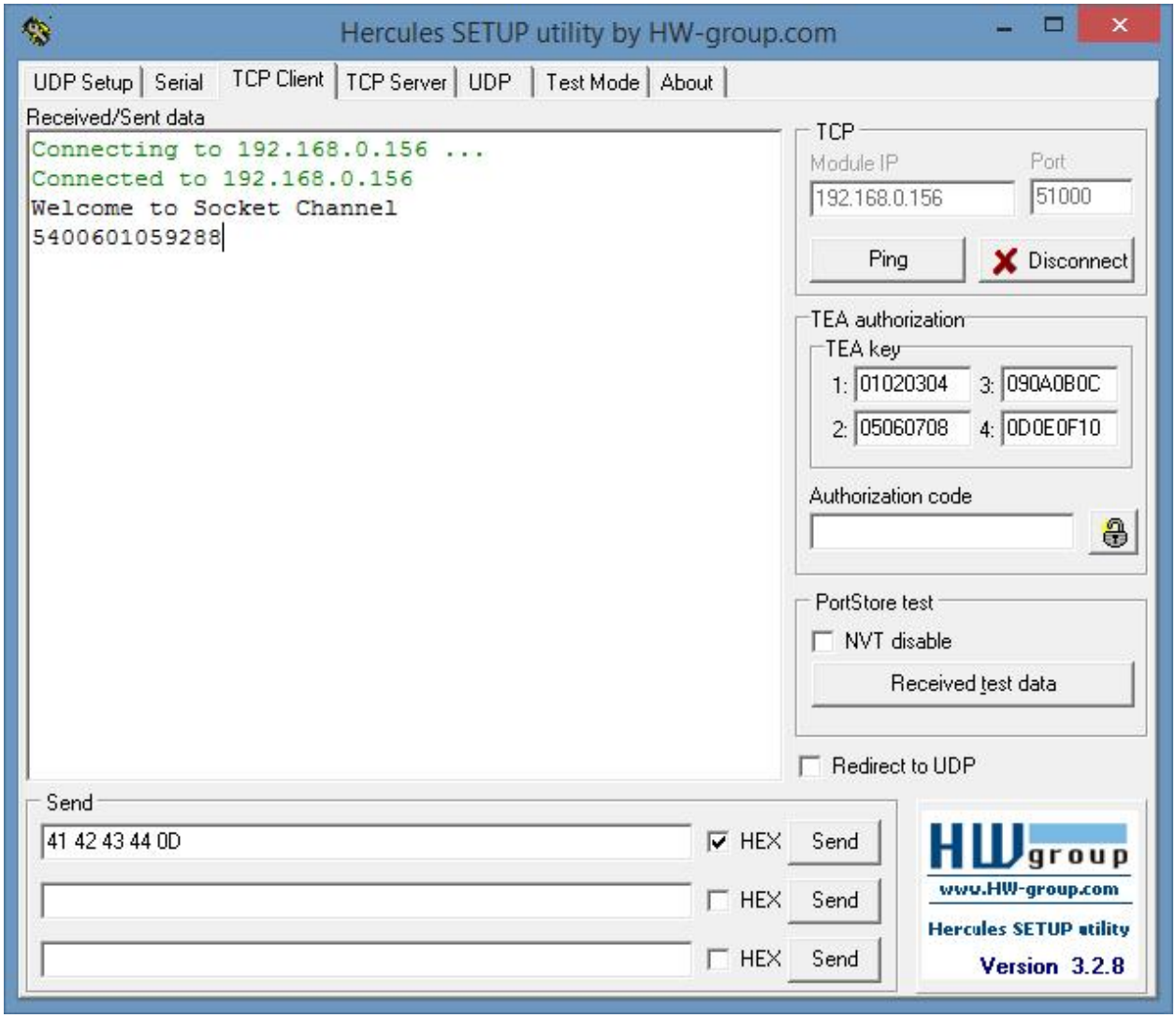
When scanning an item with RS232 or USB-COM connection, the data will be visible in ComTestSerial.





For the Ethernet setup, you need to start up Aladdin first to find the IP address of the cradle. Per default, the Ethernet cradle is setup to get the IP address via a DHCP server. Start Aladdin; select 'Device Auto detection' → 'Search on connection'. After a few seconds the Ethernet cradle will appear, and also it's IP address. In the below image, the IP address is 192.168.0.156

For the Ethernet connection test, use Hercules. Go to the TCP Client tab, and set the IP address.
Also, set the port to 51000 and press the connect button. Then scan a barcode, the data should be visible.



At this point, the scanner and cradle are setup in normal 1 way communication.
Next step is to enable 2 way and do tests.

Secondly Simple 2- way test:

First Setup the scanner into 2 way. This is done by scanning the next 2 barcodes

a) Ignore Host commands (CIFIH01)

b) Transmission mode: Ack from host (CSTTX02)

c) Ethernet optional: Active Service Selection: Both Telnet and Data Socket (CETAS02). This is only for the Ethernet cradle and allows sending back data without sending the radio address.

d) Optional: Change the timeout to 5 seconds.

When the scanner is configured for 2 way communication, and you scan a barcode, an error beep will be heard after the timeout period. This means that the scanner didn't receive an answer during the timeout period.

From now on, you can send information back to the scanner

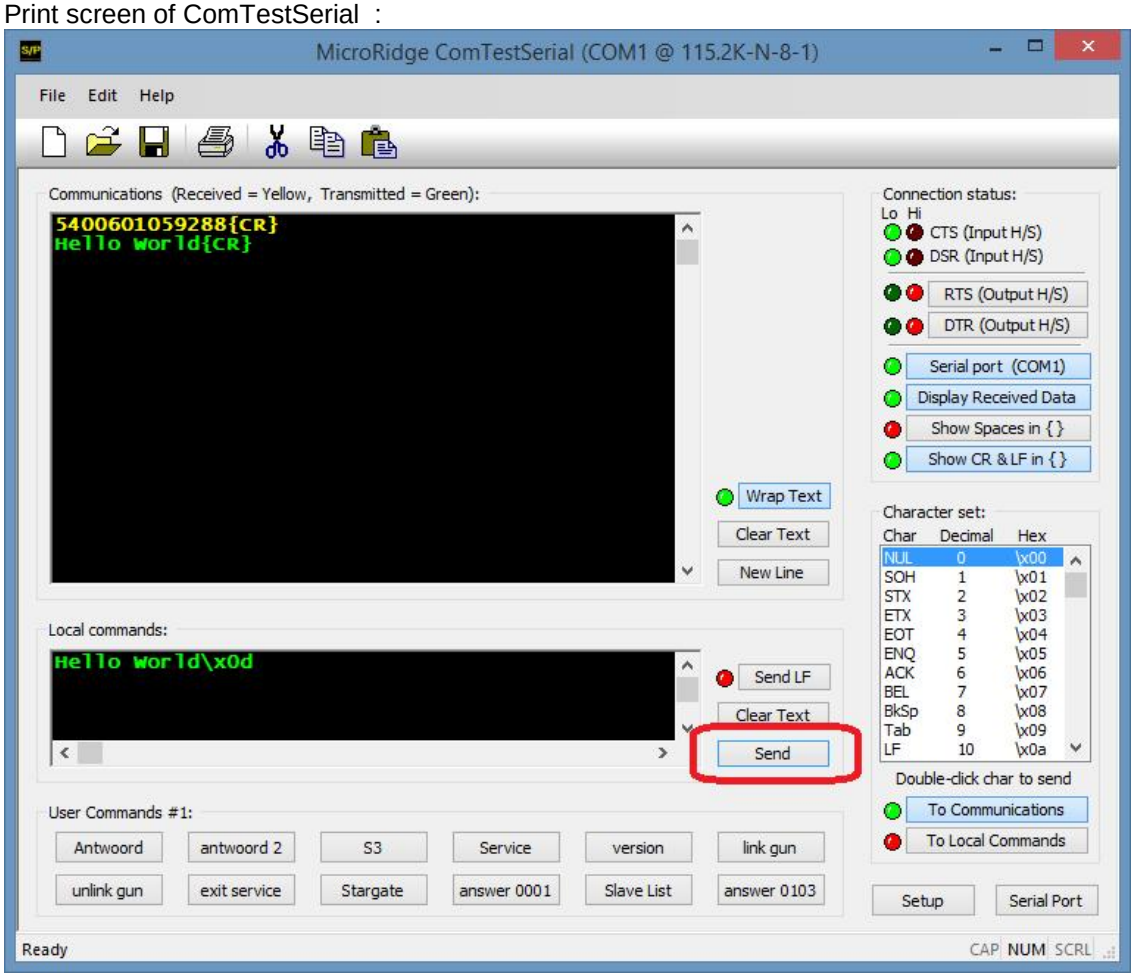
a) **Synchronous:** This means that after each message coming from the gun, the host needs to answer within the timeout.

As an example, you can send the text 'Hello world' back to the device.

In RS232 or USB-COM mode, use ComTestSerial, and paste the next text in the local commands windows. When data arrived from the scanner, press send note. The \x0d is the hex notification of the Enter character (also called carriage return). Each answer to the device must terminate with CR.

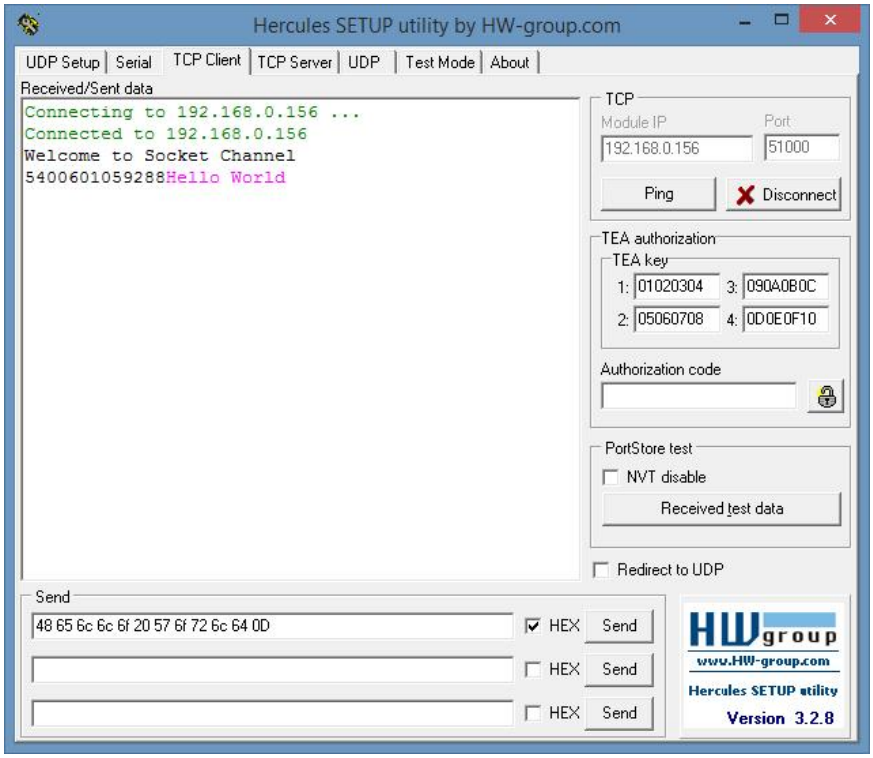
Data in to answer box: Hello World\x0d

Print screen of ComTestSerial :



In Ethernet mode, use Hercules, and paste the next text in one of the send boxes.
When data arrived from the scanner, press send. Be sure that the Hex notification is set.

Data in send box: 48 65 6c 6c 6f 20 57 6f 72 6c 64 0D

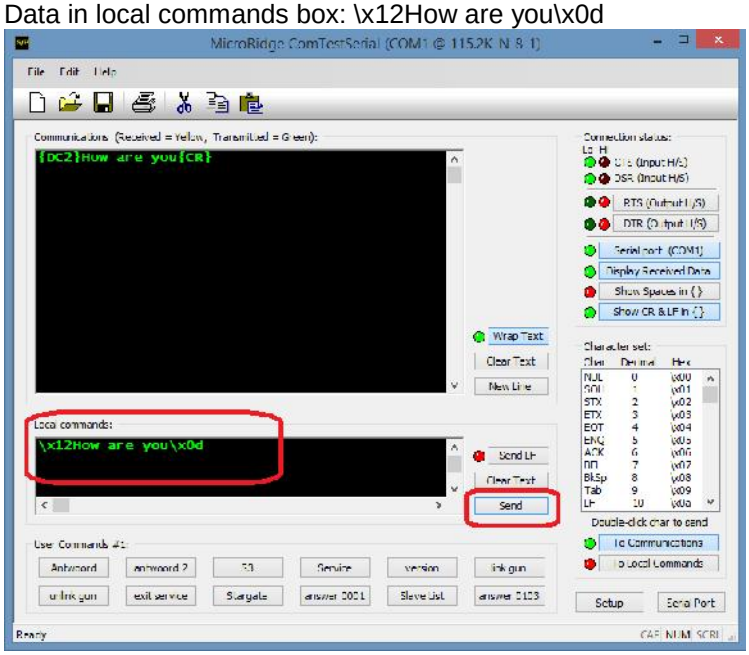


b) **Asynchronous:** This means that you can send data to the device without receiving a string from the device (also called unattended mode).

As an example, you can send the text 'How are you' to the device without scanning a barcode or pressing a key on the scanner.

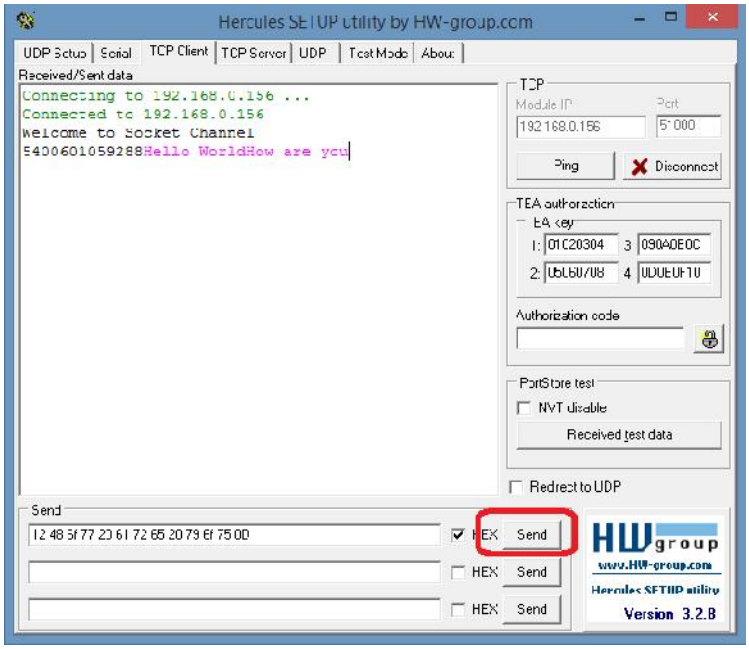
In RS232 or USB-COM mode, use ComTestSerial, and paste the next text in the local commands window. Press the send button. Note: To use the asynchronous mode, the character DC2 (hex \x12) needs to be placed in front of the message The \x0d is the hex notification of the Enter character (also called carriage return). Each answer to the device must terminate with a CR.

Data in local commands box: \x12How are you\x0d



In Ethernet mode, use Hercules, and paste the next text in one of the send boxes. Press the send button. Be sure that the Hex notification is set.

Data in send box: 12 48 6f 77 20 61 72 65 20 79 6f 75 0D



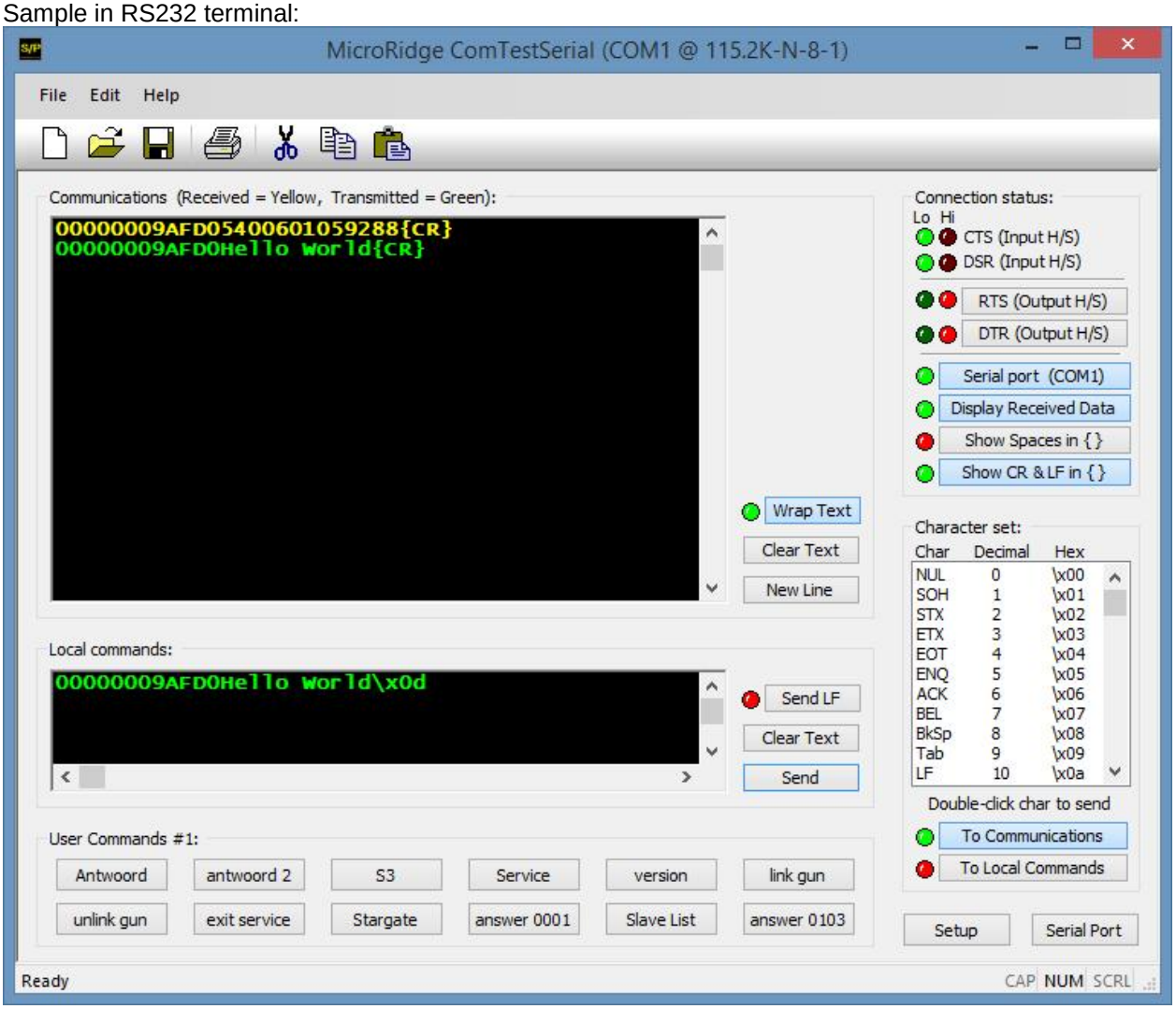
Third Optional Settings:

When working with multiple guns per cradle, it is necessary to know the radio address of the gun that is sending the message, or that you want to send an asynchronous message to. Each PM9500 has a unique 12 digit radio address. To send this radio address in front of the message, scan the barcode below.

Include Source Radio ID as prefix: (CBTAS01)

If you scan a barcode now, the cradle will put the unique radio address in front of the data.
To send back messages, you need to include this unique radio address into the answer.

Sample in RS232 terminal:



To send an asynchronous message to the gun, use the next syntax:
In the below message, the radio address is 00000009AFD0

00000009AFD0\x12How are you\x0d

What about beeps and LED's:

In the reference manual of the PowerScan 9500, you can find a section called 'Message formatting'. This section describes the Escape codes to be sent to drive the cursor, beeper and LED's.

Cursor Control

ESC [*n* A	Up **n** rows, no scroll
ESC [*n* B	Down **n** rows, no scroll
ESC [*n* C	Right **n** columns
ESC [*n* D	Left **n** columns
ESC [G	CR
ESC [*r* ; *c* H	Move to row **r**, column **c** (ESC[1;1H is the upper left character position of the display)
ESC D	Down 1 row, with scroll
ESC E	CR and cursor down 1 row with scroll
ESC M	Up 1 row and scroll

Font Selection

ESC [0 m	Normal mode
ESC [7 m	Reverse mode
ESC # 4	Large font: subsequent characters are written on the current row and the row below it using the 12x16 font which allows for two rows of eight characters on the display.
ESC # 5	Normal font: subsequent characters are written using the 6x8 font, which allows for four rows of sixteen characters on the display.
ESC # 7	Medium font: subsequent characters are written using the 8x8 font, which allows for four rows of twelve characters on the display.

Clearing Display

ESC [0 K	From cursor position to end of line inclusive
ESC [1 K	From beginning of line to cursor position (not inclusive)
ESC [2 K	Entire line
ESC [0 J	From cursor position to end of display inclusive
ESC [1 J	From beginning of display to cursor position (not inclusive)
ESC [2 J	Entire display; moves cursor to upper left corner on display

LED and Beeper Control

ESC [0 q	Emit short High tone + short delay
ESC [1 q	Emit short Low tone + short delay
ESC [2 q	Emit long Low tone + short delay
ESC [3 q	Emit good read tone
ESC [4 q	Emit bad tx tone
ESC [5 q	Wait 100 ms
ESC [6 q	Turn on the green LED
ESC [7 q	Turn off the green LED
ESC [8 q	Turn on the red LED
ESC [9 q	Turn off the red LED

The LED control escape sequences are intended to activate the LEDs for short periods of time and can be used in combination with the Beeper. The LED and Beeper will be controlled by the system after the entire command sequence is interpreted.

Example:

ESC [6 q ESC [3 q ESC [7 q	Turns on the green LED, emits a good read tone, and turns off the green LED.
ESC [6 q ESC [5 q ESC [7 q	Turns on the green LED for 100 ms and then turns off the green LED.

Setting RTC

ESC [0 p *ddmmyy*	Set date to day, month, year
ESC [1 p *hhmm*	Set time to hours, minutes; seconds are automatically set to 00.

As an example the below characters will send in asynchronous mode:

Send low tone + short delay 3 times, clear the green LED and set the red LED on.

Send data for ComTestSerial: (do not forget to change the radio address)

00000009AFD0\x12\x1b[2q\x1b[2q\x1b[2q\x1b[7q\x1b[8q\x0d